

TECHNICAL DESIGN DOCUMENT - TECHNICAL DESIGN SPECIFICATION

Project Name : Test Project
Module / Work : Technical Design
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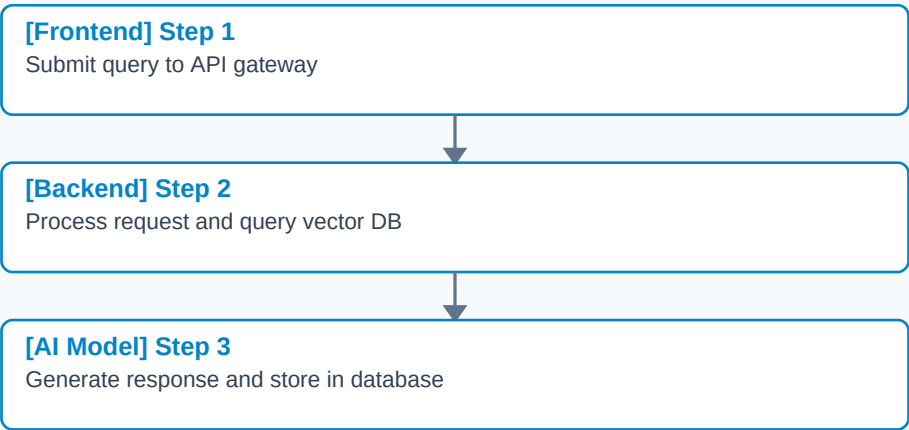
1. Technical Overview

Technical Objective::

Define system technical architecture and data flow.
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2. Architecture & Component Overview

End-to-End Processing Architecture



3. Architecture Processing Steps

Step	Component	Technical Action & Processing
1	Frontend	Submit query to API gateway
2	Backend	Process request and query vector DB
3	AI Model	Generate response and store in database

4. Component-Level Design

4.1 Feed Ingestion Component

Responsibility: Receive and process incoming daily data feeds.

Key Processing Logic::

- Parse incoming feed file.
- For each account/record in the feed:
 - If account balance is negative -> create new Overdraft record.
 - If account has un-invested cash -> create new Cash record.

5. API Specifications & System Integrations

API Name	Method	Endpoint	Purpose	Status
Upload BRD	POST	/api/documents/upload	Upload BRD file	200 OK

6. Data Model & Database Design

6.1 Table: Document

Field / Column	Data Type	Key	Required	Description / Constraints
id	UUID	-	Yes	PK

7. End-to-End Data Flow

End-to-End Data Flow Sequence

[Upstream Feed -> Feed Handler] Step 1

Transmit daily settlement file

Step	Source System	Target System	Action Performed	Payload / Transferred
1	Upstream Feed	Feed Handler	Transmit daily settlement file	CSV Record Feed

8. Security Design & Controls

ID	Security Area	Security Control Specification	Source Standard
SEC-01	SSO Authentication	SAML 2.0 / OAuth2 with JWT validation	Enterprise Security

9. Validation Rules

Frontend Validation Rules

Field	Validation Rule	User Error Message
accountNumber	Must be 10 numeric digits	Invalid Account Number

Backend Server-Side Validation

Field / Parameter	Validation Rule	API Error Response
amount	Must be non-negative float	Invalid Amount

Database Constraints

Constraint Name	Description
UQACCOUNTNUM	Unique index on account number

10. Error Handling & System Resilience

Error Scenario	Layer	Handling Logic	Response	User Experience
Feed Timeout	Ingestion Layer	Retry with backoff	HTTP 504	Alert banner

11. Edge Cases & Technical Mitigations

Scenario	Impact	Technical Mitigation & Handling
Negative Cash Balance with Missing Profile	High	Quarantine record to Exception Queue

12. Hard Technical Constraints

ID	Hard Constraint Specification	Source / Rule
C-01	Aging must be calculated strictly in calendar days	BRD Section 4

13. Open Technical Questions & Clarifications

ID	Technical Question / Unknown	Architecture Impact	Source
OQ-01	Confirm feed delivery mechanism	Ingestion Architecture	BRD Review